M359

Conceptual Data Model Comparisons

 WEB 98690 0
1.1

Conceptual data model comparisons

There are many different varieties of ER diagrams all of which have similarities. Here we will look at some of the common versions and compare them with the Open University's preferred method. It needs to be noted that all conceptual data models are attempting to do the similar things and it does not matter which method is used it is the concepts behind the models that matter. Once the concepts have been learned the translation from one method to another is simple.

This document will compare different notations with the Open University's preferred method by using the same conceptual data model in each instance. The other notations being compared are Chen, UML, IDEF1X, SSADM, and Information Engineering Model.

Note:

The example scenario and models given in this document are not related to those in the course texts, they are used only within this document.

The examples of the various modelling notations are not intended to be exhaustive but are for comparison with the notation used on the OU courses; there are features of the various notations that are not presented here.

The additional model notations presented here are not assessed in the M359 course.

If you identify any mistakes or ambiguities in this comparison the course team would be delighted to hear from you, similarly if you want to suggest additional notations that could be compared with the one used on M359 then do drop us a line.

Acknowledgement

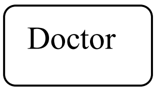
The course team would like to thank Derek Richardson for his work on the original version of this document.

Open University course notation

NOTE: This is the method used by the Open University database courses. This is the notation taught and is the notation that you will use during the course.

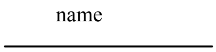
The Open University uses crow’s foot notation in which entities are joined by a relationship, which may be optional or mandatory, one to one, one to many or many to many and must be named.

Entity types appear in round cornered boxes

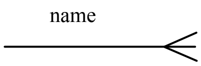


Relationships are named and can be

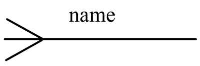
one to one 1 : 1



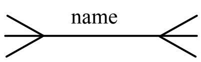
one to many 1 : m



many to one m : 1



many to many m : n



Open circle optional ○

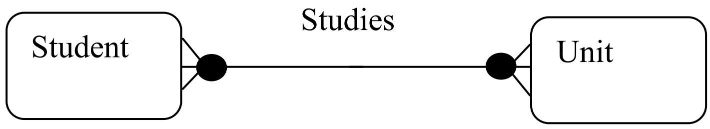
Solid circle mandatory ●

In the M359 course notation we use only binary relationships, that is a relationship is between two entity types (or one entity type twice if it is recursive!). Other notations allow the representation of n-ary relationships involving more than 2 entity types. For example, UML uses a diamond to represent such relationships. N-ary relationships can be decomposed into binary relationships and additional constraints. However, we do not address this in the M359 course. We will not give examples of n-ary notations in this comparison.

Attributes belonging to each entity are listed at the end in the form: Doctor (StaffID, Name, Specialism, Position). The identifying attribute(s) are underlined.

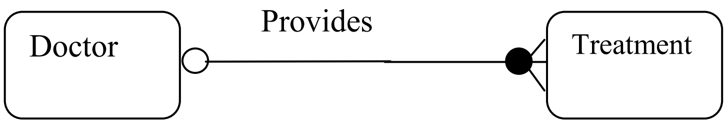
For example:

Student and **Unit** entity types can be connected by the **Studies** relationship as



A Student studies 1 or many Units →
A Unit is studied by 1 or many Students ←

A Ward is staffed by one or many Nurses →
A Nurse staffs exactly one Ward ←



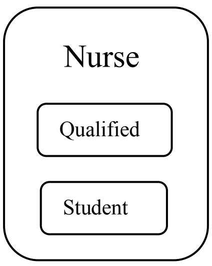
A Doctor provides zero, one or many Treatments →
A Treatment is provided by exactly one Doctor ←

Each entity type must also have its attribute(s) defined with the identifying attribute underlined:

Doctor (StaffId, Name, Specialism, Position)

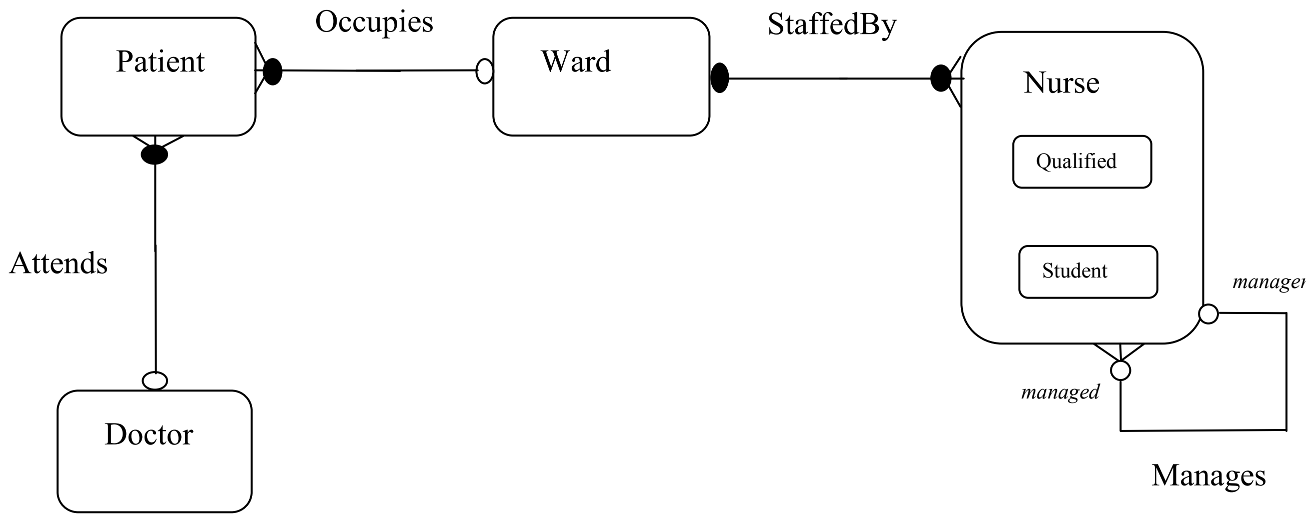
Super types and sub types

Sub types are represented as entities within their super type entity type.



Simple hospital scenario in Open University notation

Entity relationship diagram



Entities	**Attributes**
Doctor	(<u>StaffId</u>, Name, Specialism, Position)
Patient	(<u>PatientId</u>, Name, Address, DateAdmitted)
Ward	(<u>WardID</u>, WardName)
Nurse	(<u>NurseId</u>, NurseName, Grade)

Qualified (QualfiedDate)

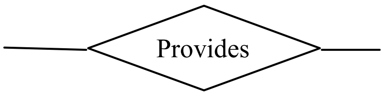
Student (CommencedStudyDate)

Chen notation

Peter Chen invented entity / relationship modelling in the 1970's and it is still used today. The basic components are

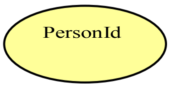
Entity type

Weak entity type



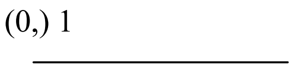
Relationship

Weak Relationship

Attributes 

these are connected to their Entity type by lines.

Cardinality (placed alongside the line connecting the relationships to the entity types)

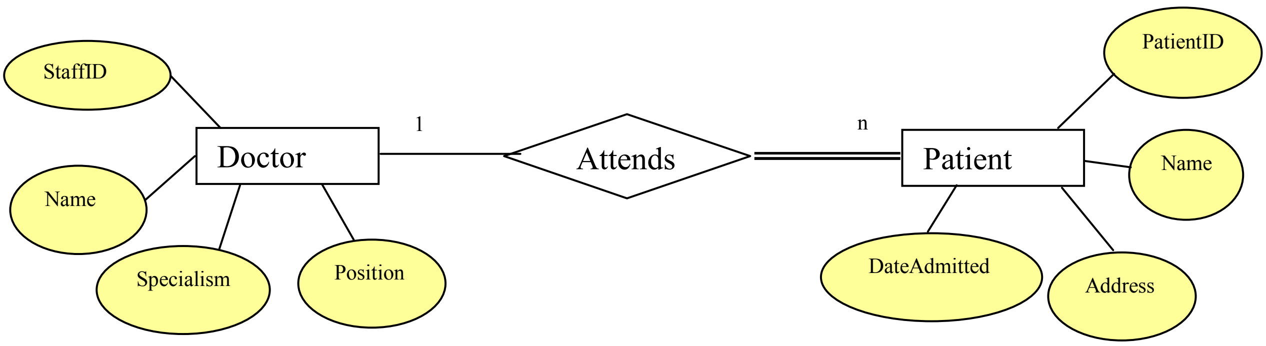


optional (single line) zero or one

mandatory (double line) 1 or many

Some versions of Chen notation do not use the double line notation for mandatory participation but just use a single line and the fact that the first number in the cardinality is a one makes the participation condition mandatory, other notations drop the first number in the cardinality and use the single/double line notation instead.

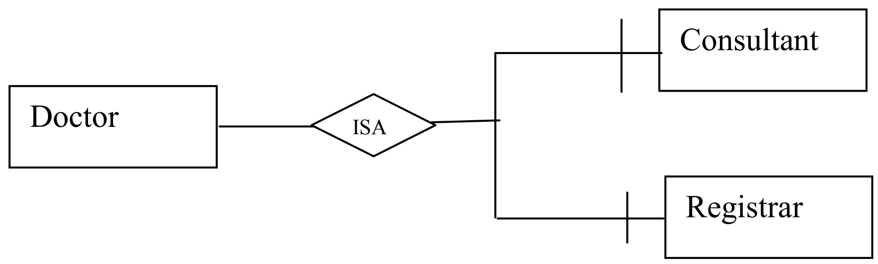
For example:



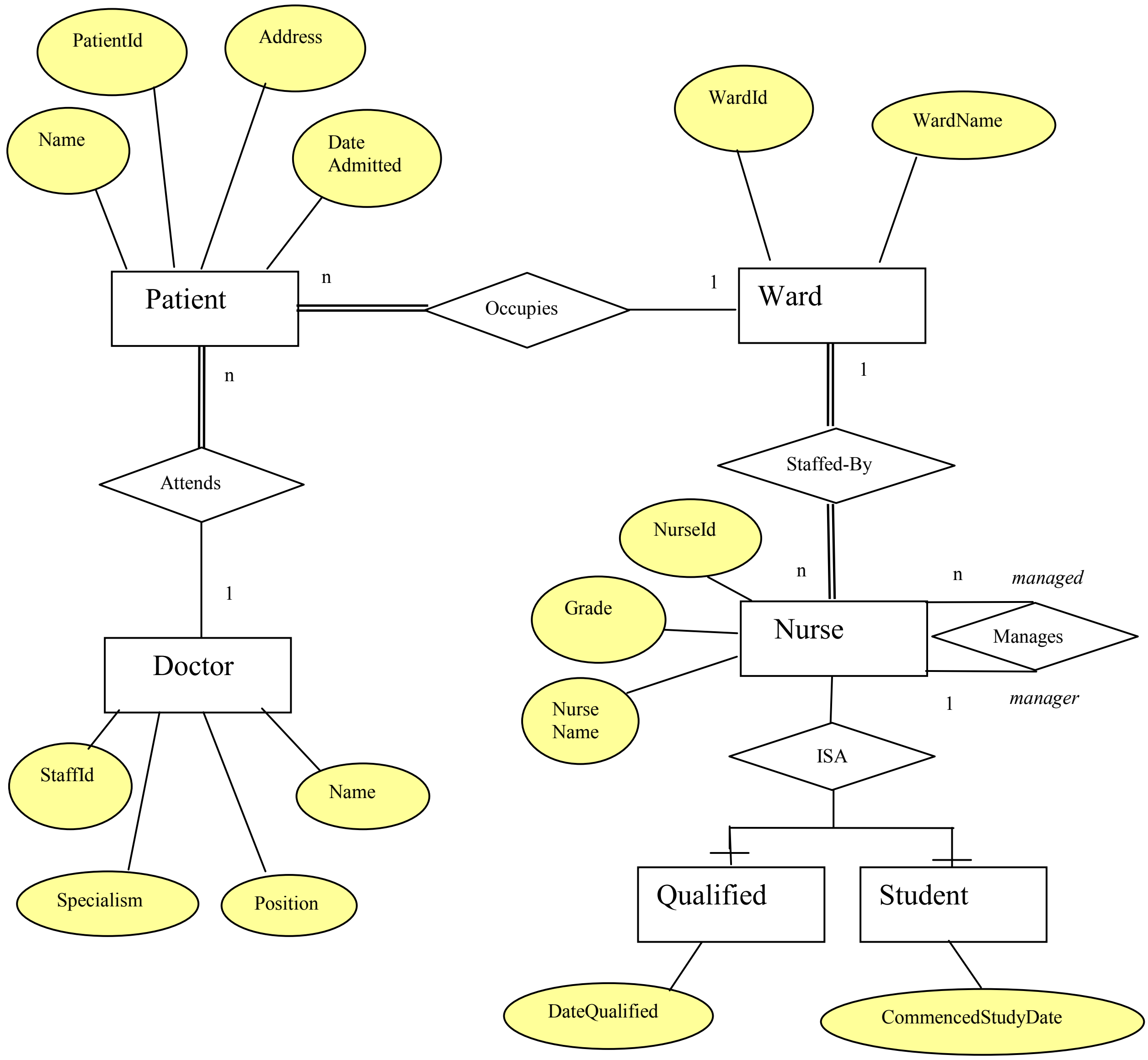
The doctor attends the patient is optional but patient is attended by the doctor has mandatory participation.

Super and sub types

Chen uses the ISA relationship with the addition of a barred line to show an either or option (exclusive) relationships. So, in the following example a doctor is either a consultant or a registrar, but never both.



The simple hospital scenario in Chen notation

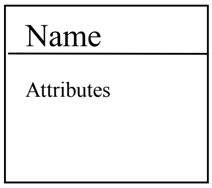


Unified Modelling Language (UML)

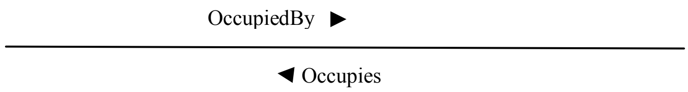
UML is not classified as a data modelling language but as an object modelling language but on close inspection it looks very much like an entity relationship diagram. Ivar Jacobson [Jacobson 1992, p. 132] called these classes entity objects. UML includes a much richer set of features than the notation we use, for example it has two distinct forms of relationship – the association relationship which is comparable to the OU relationship as it represents associations between instances, and generalisation relationships which are comparable to the OU sub typing as they represent an IS-A relationship between type.

The **entity type** (object class) representation consists of a square sided rectangle with one or more compartments, in data modelling it is usually shown with two compartments. At the top is the class name which is the only part that must be included, this is followed by a list of attributes.

We do not consider the use of UML stereotypes that could be used to represent the OU entity relationship diagram components.



The **association relationship** is a line connecting two entity types, which can be named in *both* directions



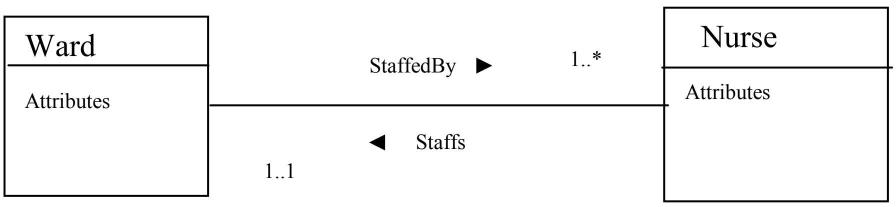
The **cardinality** and **optionality** are shown by the numbers at each end of the line

Using a notation that shows the upper and lower limit of the cardinality. So, in 0..* the lower limit denotes optionality (0 = optional) and the upper limit the cardinality (* = one or many). And, 1..1 would mean exactly one (between the lower limit of 1 and the upper limit of 1) which is commonly written as 1.

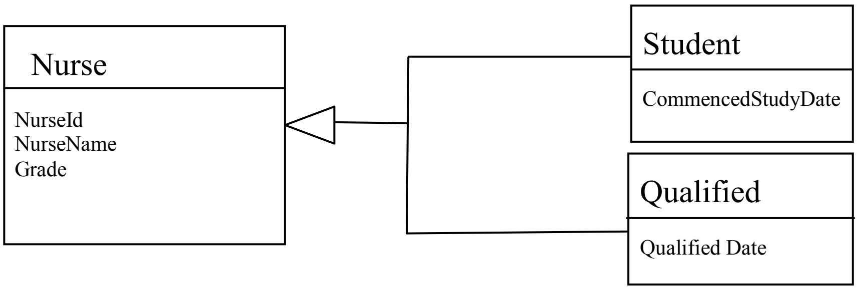
For example

A Ward is staffed by one or more Nurses.

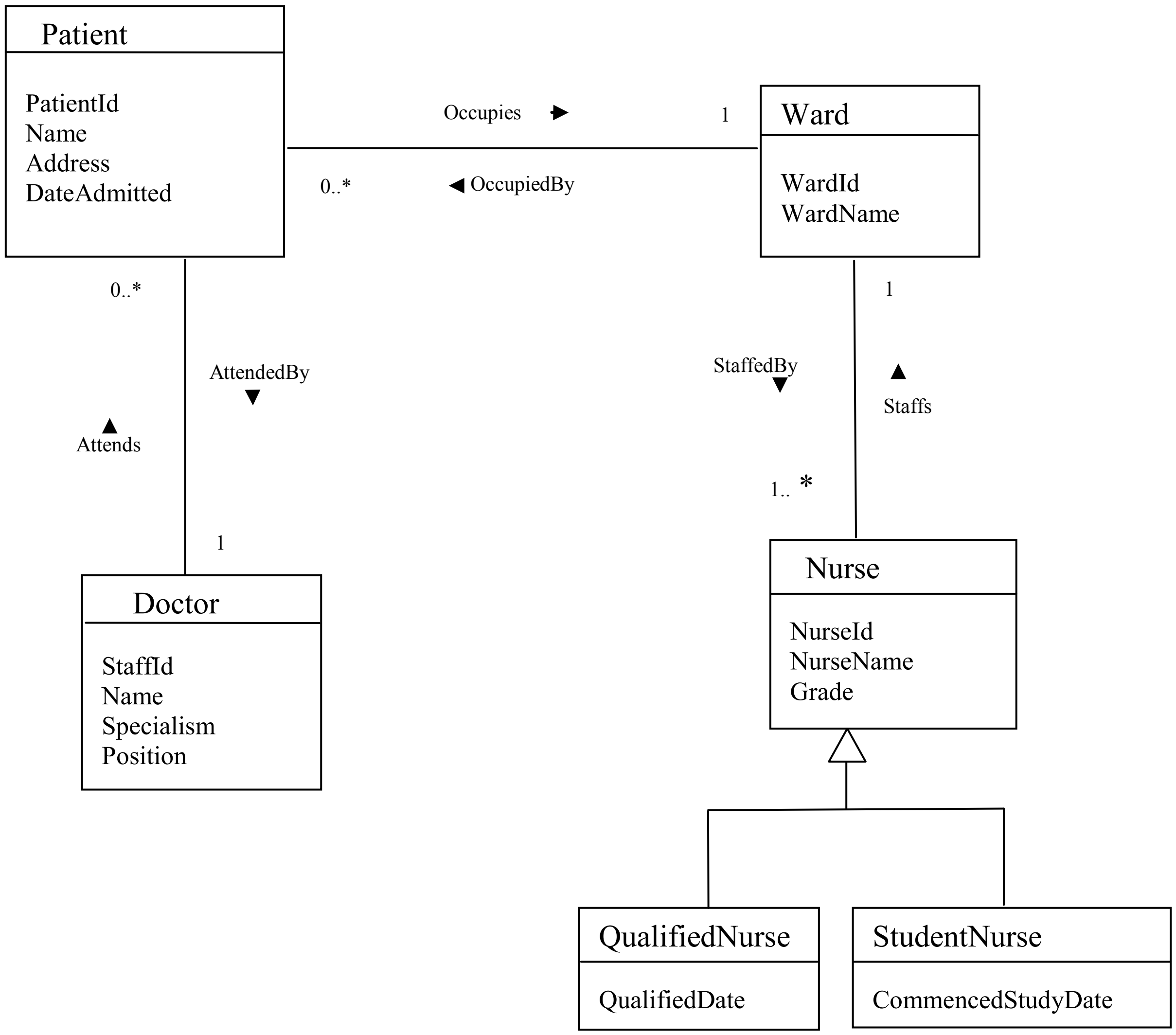
A Nurse staffs exactly one Ward.



Sub Types are shown by generalisation relationships. In effect these show relationships between types rather than between instances of those types. So here the Nurse type is a generalisation of a Student type and a Qualified type.



The simple hospital scenario in UML



IDEF1X

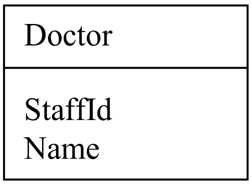
IDEF1X is a design notation (rather then a modelling notation) that is used in many branches of the US federal government and has quite complex variations within the notation. Design notations include details such as date type for attributes, which do no appear in our conceptual modelling notation.

Entity type

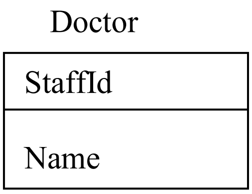
Dependent entity type

A dependent entity type is one in which the primary key of the entity type contains two attributes one of which is a foreign key in another entity type (comparable to the weak entity type used elsewhere).

Attributes can be listed with the entity type in a manner similar to UML.



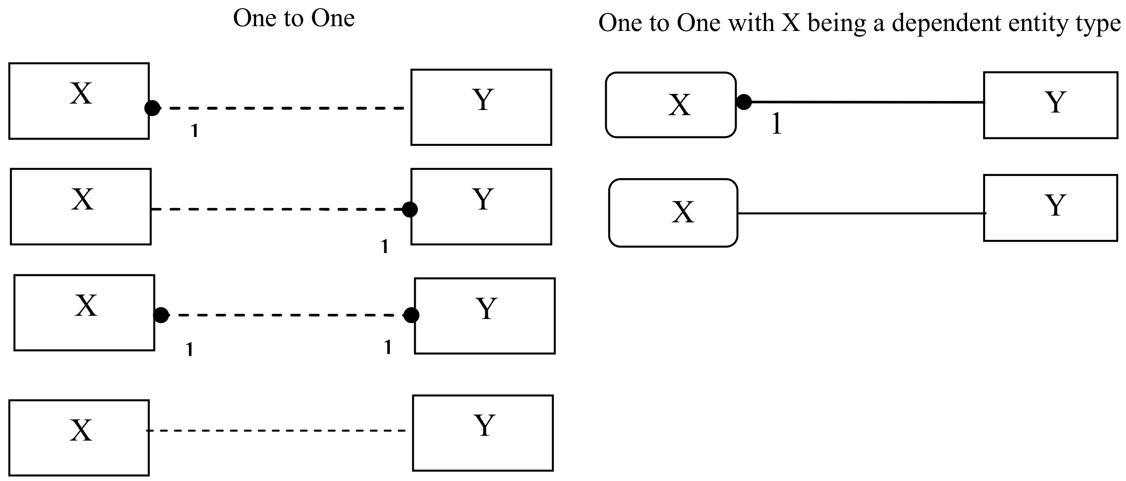
Sometimes the entity type name is shown above the entity type box, with the identifier attributes separated from the other attributes.



Relationships

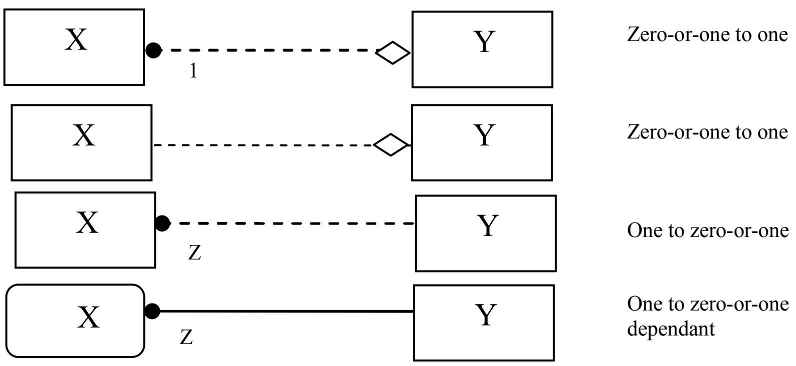
Idef1x uses a rather complex notation for relationships. Unfortunately, Idef1x models are not easy to understand as they can use a variety of symbols to represent the same thing, which will vary as a result their context within the model. Degree and optionality notations change depending on whether or not the entity type is dependent (comparable to weak entity types). If the unique identifier of the entity contains a foreign key attribute belonging to the other entity in the relationship the entity box changes to round corners and the dashed relationship line becomes solid.

For example here are six ways of displaying a one to one relationship



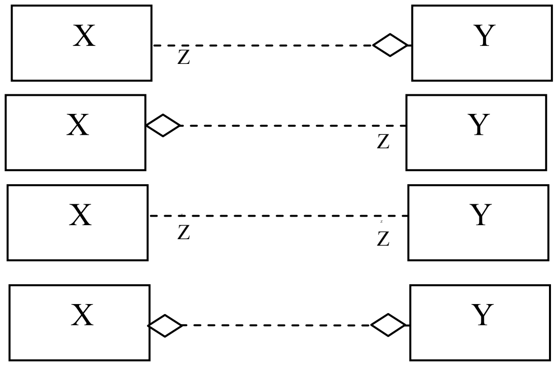
These six variations all represent, one to one with mandatory participation at both ends.

There are then four variations that have **optional** on one side

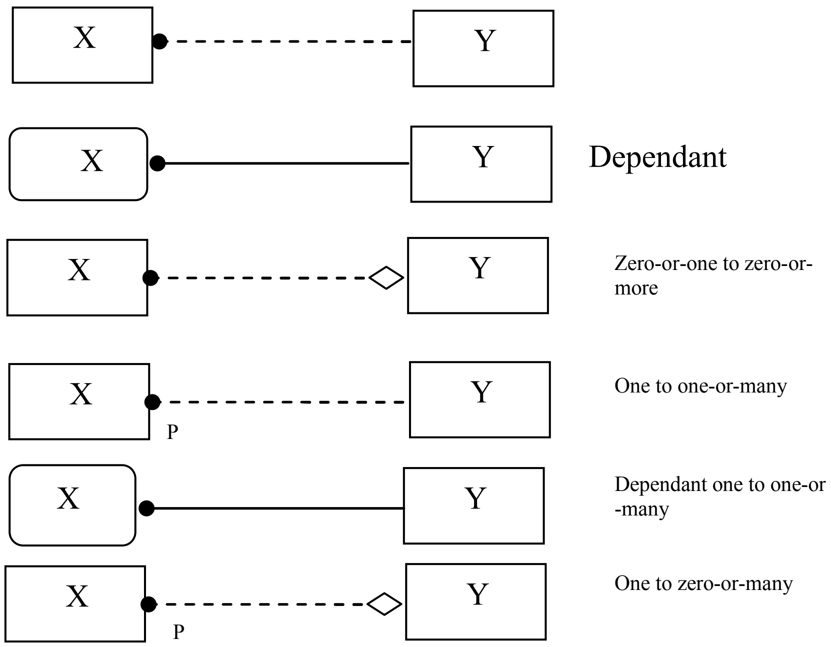


There are another four variations that are optional on both sides

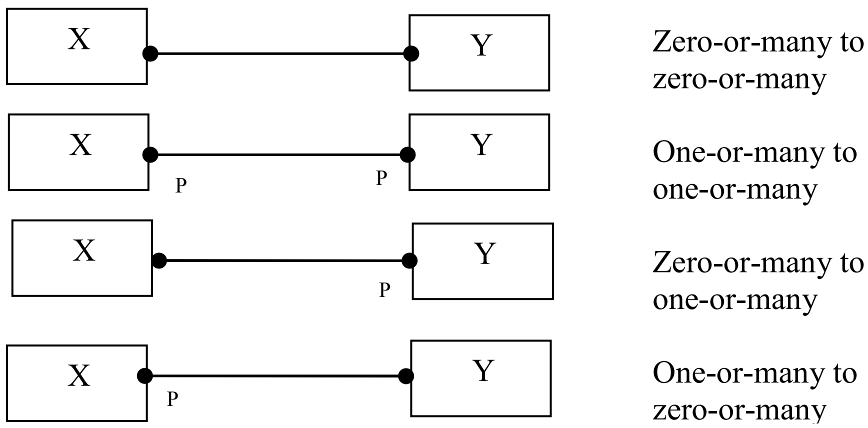
That is Zero-or-One to Zero-or-One



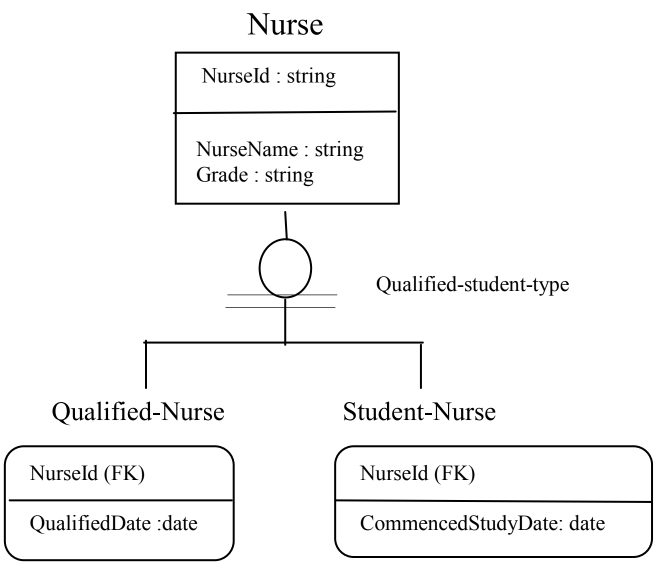
One(Y) to zero or more (X)



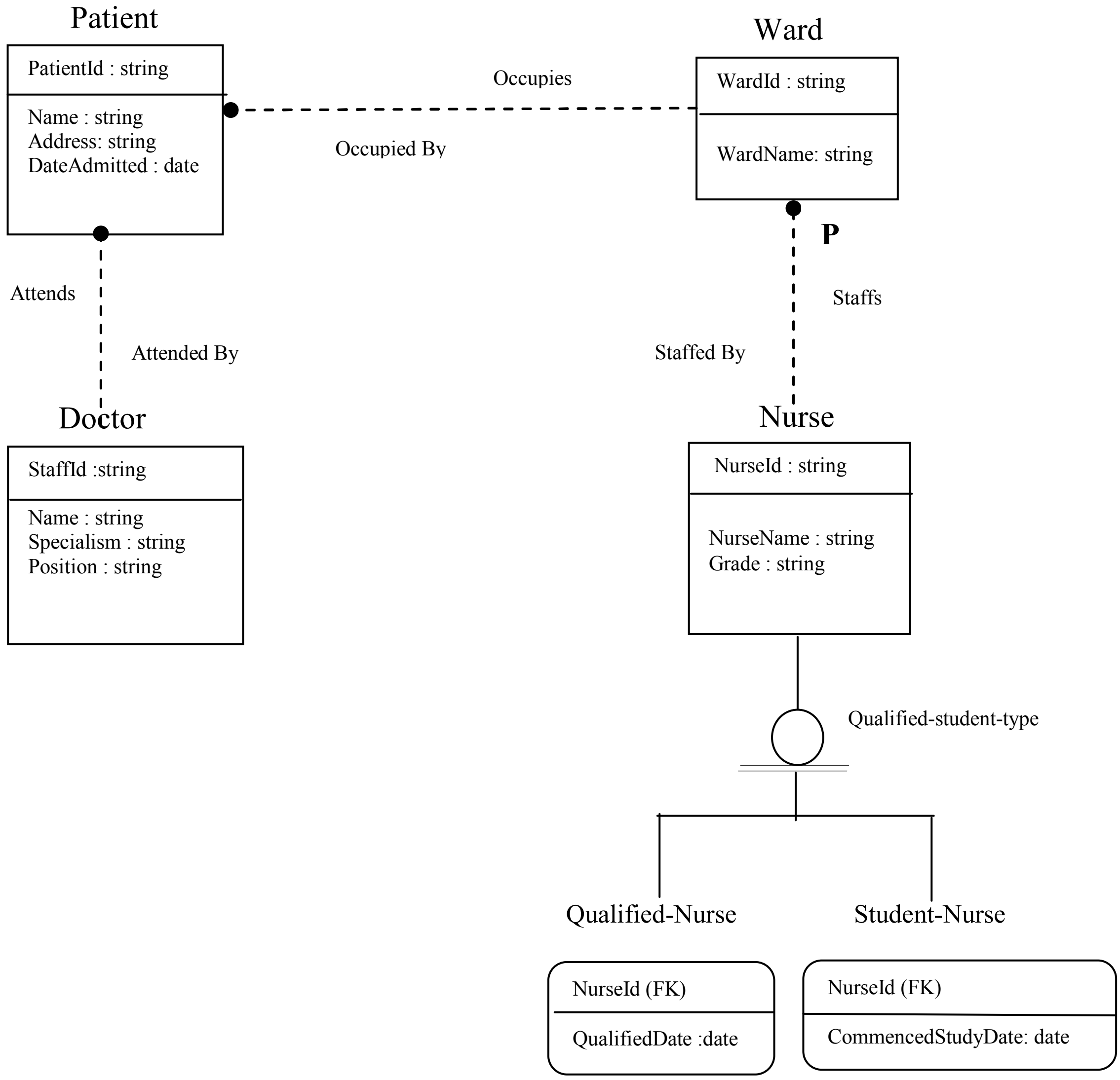
Finally the **many to many variations**



Sub Types are shown by an isa relationship The circle with the double line indicates that all occurrences of the sub type must belong to the super type (complete). A circle with a single line indicates an incomplete sub type in which the sub types do not represent all possible occurrences of the super type. So, if the double line shown below were a single line there could be Nurses who were neither students nor qualified.



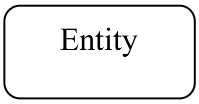
The simple hospital scenario in IDEF1X



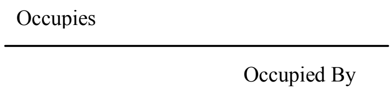
SSADM

In SSADM entity relationship modelling are called logical data models

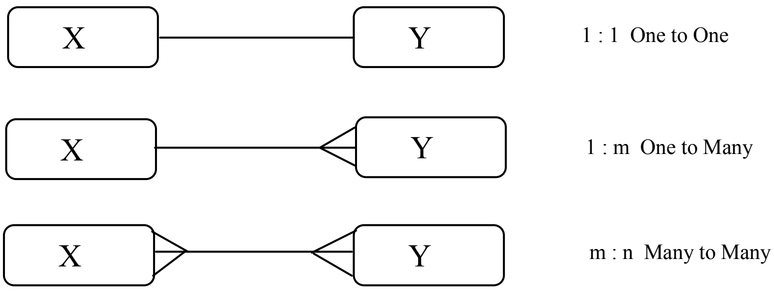
Entity type a box with rounded corners.



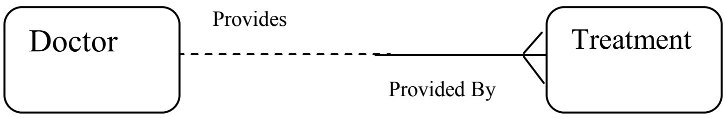
Relationship is a line connecting two entities which is named in both directions.



With SSADM there are three possible types of degree notation



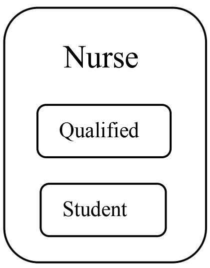
Mandatory and optional occurrences are defined by the use of a dashed line for the optional involvement in relationships



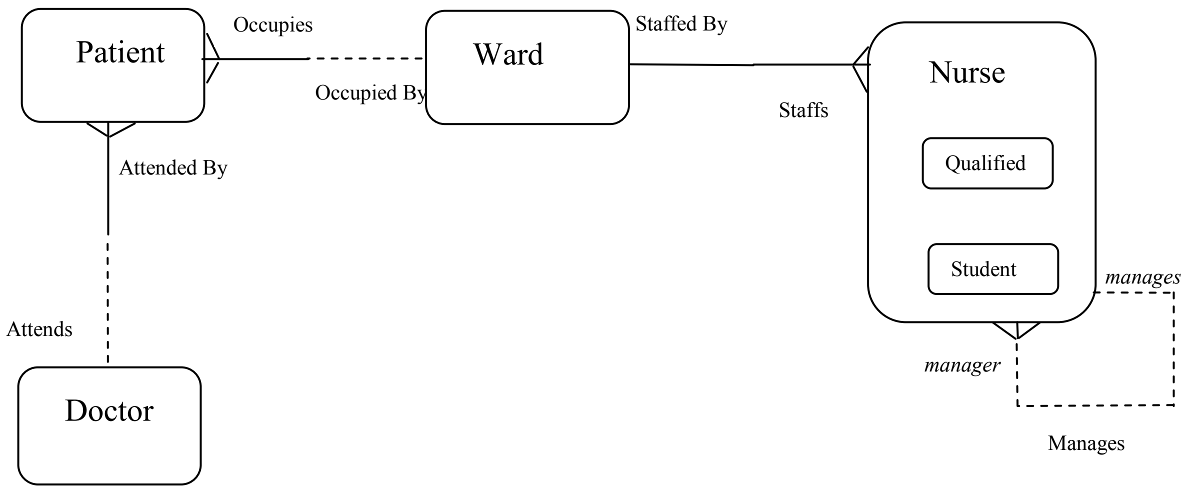
A Doctor provides zero, one or many Treatments →
A Treatment is provided by exactly one Doctor ←

Super and sub types

Sub types are drawn as entity types within entity types



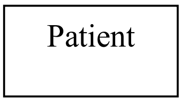
The simple hospital scenario in SSADM



Information engineering model

The information engineering model was developed by Clive Finkelstein, in the late seventies, in Australia. He worked together James Martin to publicise it in the USA and Europe. Over time they produced two versions which also have minor variations.

Entities are named square boxes



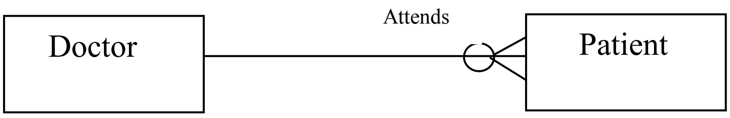
Attributes are not shown in the ERD

Relationships are shown as solid lines

Cardinality / optionally

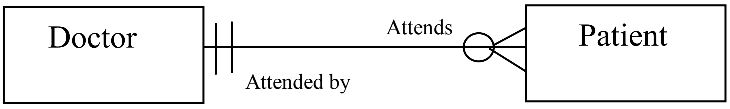
Optionality is shown by the addition of an open circle. If an occurrence of the first entity may or may not be related to occurrences of the second then the open circle appears near the second entity. (Note this is the opposite end to the one you would expect with the M359 course notation.)

A crow's foot denotes a 'many' relationship and can appear either with an open circle (showing optional) or a single bar across the relationship line (showing mandatory).



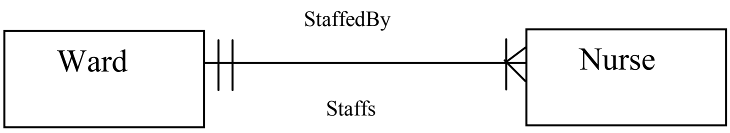
A Doctor attends zero, one or many Patients

If a relationship involves only one occurrence of an entity type then a double bar crosses the relationship line



A Patient is attended by exactly one Doctor

To show a cardinality of 1 or more a single bar is added to the crow's foot.

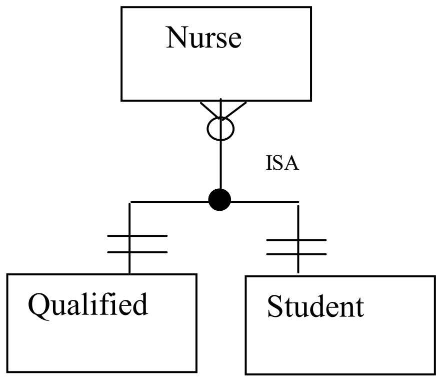


A Ward is staffed by one or more Nurses.

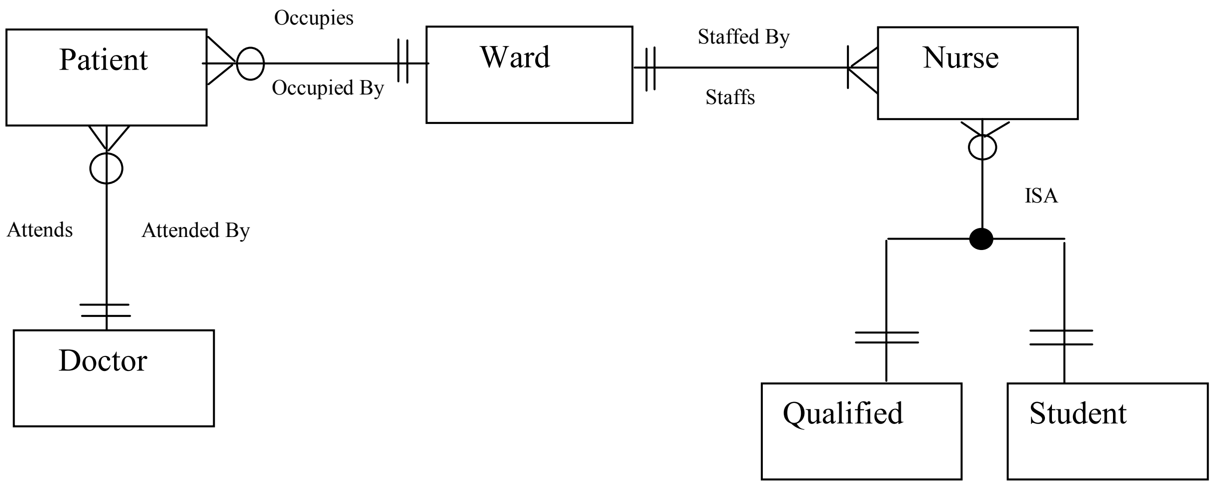
A Nurse staffs exactly one Ward.

To show 0 or 1 it would be an open circle and a single short line.

Sub types are shown as external boxes to the super type and are joinbed by an ISA relationship. The double crossed line at the sub type means that the nurse must be either or of the Qualified or Student Nurse



The simple hospital scenario using the information engineering model



Summary

As can be seen from the above methods each has its own way of describing a given scenario, they are all similar and have varying uses depending on the type of modelling that is being undertaken.

Chen was the first and so is not as complex as some of the later modelling techniques.

UML is not strictly a conceptual data modelling language it is an object modelling language. Other entity relationship diagrams allow the user to see at a glance the relationships between entities but in UML you have to read the diagram carefully as it contains much more information than is normal for an ERD.

IDEF1X, a design notation rather than a modelling notation, is very difficult to learn as can be seen from the description contained here; there are a variety of different symbols that can mean the same thing depending on the circumstances in which they are used.

SSADM is a simple conceptual data modelling technique which has been used by the UK government as a standard modelling technique.

The Information Engineering Model has two versions with slight differences but is an easy to use method

These five notations are only a few of many notations in use and this comparison is meant only to be a general overview; it is neither complete nor definitive. There are many other conceptual data modelling techniques around and if you would like us to present a comparison with any other techniques then get in contact with the course team, or supply an example of the simple hospital scenario to show how its notation appears.

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